

# Secure Coding Review

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**DOMAIN:** Cyber Security

## INTRODUCTION

Secure Coding Review is the process of examining source code to identify security vulnerabilities, coding errors, and weaknesses that could be exploited by attackers. It helps developers build reliable and secure software by following secure coding practices.

## OBJECTIVE

- Identify security vulnerabilities in code.
- Detect insecure coding practices.
- Improve application security.
- Ensure compliance with coding standards.

## TOOLS USED

- Python 3
- IDLE Python
- Windows Operating System
- SQL Query Demonstration

## PROGRAM CODE

```
username = input("Enter Username: ")  
password = input("Enter Password: ")
```

```
query = "SELECT * FROM users WHERE username='" + username + "' AND password='" +  
password + "'"
```

```
print(query)
```

# OUTPUT SCREENSHOTS

```
task 3.py - C:/Users/DELL/Desktop/task 3.py (3.14.6)
File Edit Format Run Options Window Help

username = input("Enter Username: ")
password = input("Enter Password: ")

query = "SELECT * FROM users WHERE username='" + username + "' AND password='" + password + "'"

print("\nGenerated SQL Query:")
print(query)
|
```

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```
"IDLE Shell 3.14.6"
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/DELL/Desktop/task 3.py =====
Enter Username: RAMYA
Enter Password: 1234

Generated SQL Query:
SELECT * FROM users WHERE username='RAMYA' AND password='1234'
>>>
```

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## **FEATURES**

- Reviews source code for vulnerabilities.
- Detects SQL Injection risks.
- Improves software security.
- Supports secure development practices.

## **ADVANTAGES**

- Enhances application security.
- Reduces security risks.
- Improves code quality.
- Helps prevent cyber attacks.

## **APPLICATIONS**

- Secure Software Development
- Cybersecurity Audits
- Vulnerability Assessment
- Code Quality Review
- Educational Projects

## **CONCLUSION**

The Secure Coding Review project demonstrates how insecure coding practices, such as directly concatenating user input into SQL queries, can introduce security vulnerabilities like SQL Injection. Reviewing and improving code using secure coding techniques helps build safer and more reliable applications.